BOMI KWON

Seoul, Republic of Korea

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Research Interests: Structured and controllable generative modeling, diffusion models, and text-conditioned generation.

EDUCATION

Sookmyung Women's University Seoul, Republic of Korea
Master of Science, Department of Computer Science Mar. 2025 – Feb. 2027 (expected)

- Current GPA of 4.41/4.5

Bachelor of Science, Department of Statistics Mar. 2020 – Feb. 2025
Bachelor of Business Administration, Department of Business Administration

- Total GPA of 3.61/4.5

PUBLICATIONS

B. Kwon and K. Y. Lee (2026). **Text-Conditioned Scene Graph Sequence Generation via Discrete Diffusion.** *Proceedings of the 35th ACM International Conference on Information and Knowledge Management (CIKM '26).* Accepted Short Paper.

B. Kwon and K. Y. Lee (2026). **Text-driven 2D Human Pose Generation via Diffusion in Directional Angle Space.** *The Transactions of the Korea Information Processing Society*, 15(7), 610–619.

HONORS & AWARDS

Selected Research Principal Investigator, 2026 WISET Engineering Research Team Program for Women Graduate Students (General Track) Apr. 2026 – Oct. 2026

- Selected as one of **50 research teams nationwide**; awarded **KRW 7,000,000** in competitive research funding.

Sookmyung Leadership Activity Scholarship Jan. 2023

PRESENTATIONS

B. Kwon and K. Y. Lee (2026). *Diffusion-based Skeleton Pose Editing.* 2026 Spring Conference of the Korean Institute of Smart Media and the Korean Society of Digital Industry.

B. Kwon and K. Y. Lee (2025). *A Graph Generation Model for Text-based Graph Editing.* Korea Software Congress 2025 (KSC 2025).

B. Kwon and K. Y. Lee (2025). *Development of a Large-Scale Graph Retrieval System based on LLM and Graph Database Providing a Natural Language Interface.* Korea Computer Congress 2025 (KCC 2025).

B. Kwon, S. Park, Y. Park, and K. Y. Lee (2026). Text-Conditioned Diffusion for Static Humanoid Robot Pose Generation with Joint-Limit Embedding and Self-Collision Guidance.
Submitted to KICS 2026.

RESEARCH EXPERIENCE

Sookmyung Women's University Seoul, Republic of Korea
Graduate Research Assistant, Data Intelligence Laboratory Mar. 2025 – Present

- Conduct research on generative AI, with a focus on diffusion models, structured generation, and text-conditioned generation.
- Design data pipelines, model architectures, experimental protocols, and evaluation frameworks for generative modeling research.
- Implement baselines, analyze experimental results, and prepare manuscripts for journal and conference submissions.

WISET Engineering Research Team Program for Women Graduate Students
Research Principal Investigator Apr. 2026 – Oct. 2026

- Lead a research team of two undergraduate students developing synthetic human-pose data generation methods for industrial safety research.
- Manage research planning, experimental direction, budget execution, and coordination with participating members and related institutions.
- Mentor undergraduate and secondary-school students in generative AI, diffusion models, dataset construction, and research methodology.

PROJECTS

Text-Conditioned Scene Graph Sequence Generation via Discrete Diffusion 2025 – 2026

- Developed a discrete diffusion framework for generating time-ordered scene graph sequences from natural-language descriptions of human–object interactions.
- Designed a Transformer-based model to capture object relationships within each frame, temporal changes across frames, and alignment with the input text; constructed training data from **Charades-STA and Action Genome** and evaluated the framework against multiple retrieval and generative baselines.
- **Led the project end-to-end as first author**, including problem formulation, model design and implementation, experimental design and analysis, and manuscript preparation.

Text-Conditioned Humanoid Robot Pose Generation with Physical Constraints 2026

- Developed a text-conditioned diffusion model for generating physically valid static poses for the **Unitree G1** humanoid robot.
- Retargeted human motion data to Unitree G1 in **MuJoCo** and incorporated joint-range and self-collision constraints into the generation process.
- Designed constraint-aware generation methods to improve physical validity while maintaining text-based pose control and diversity.

Text-driven 2D Human Pose Generation via Diffusion in Directional Angle Space 2025 – 2026

- Proposed an angle-based representation for text-driven 2D human pose generation instead of directly generating joint coordinates.
- Built a diffusion-based generation pipeline using **BGE-M3** text representations and forward kinematics for pose reconstruction.

AISUM Industry–Academic Collaboration — Large-Scale Product Matching and Retrieval System 2025 – Present

Product Matching (PM)

- Contributed to an image-based product matching system combining **object detection and visual embedding models** for similar-product retrieval.
- Evaluated multiple image embedding models and ensemble strategies and developed object-aware ranking methods for product search.

Semantic Matching (SM)

- Developed a semantic retrieval system for recommending products that are contextually relevant to the content and intent of online articles.
- Used **LLM prompt engineering, BGE-m3-ko/KURE-v1 embeddings, and Qdrant** to build large-scale product representations and semantic search pipelines.

Agentic Retrieval System

- Integrated Product Matching and Semantic Matching functions into an **MCP-based tool interface** for use with LLM clients including GPT, Gemini, and Claude.
- Implemented a **LangGraph-based agent workflow** to orchestrate tool selection and multi-stage product retrieval, including automated follow-up searches.

Development of a Large-Scale Graph Retrieval System based on LLM and Graph Database Providing a Natural Language Interface 2025

- Developed a natural-language graph retrieval system that converts user queries into **Cypher queries** using an LLM and searches large-scale graph data stored in **Neo4j**.
- Built an end-to-end pipeline for graph generation, metadata-based storage, retrieval, and real-time visualization.
- Evaluated the system using natural-language queries with varying levels of complexity and ambiguity.

ENTREPRENEURIAL EXPERIENCE

fitapat (ZOOC)
Founding Team Member / Product & Marketing Lead Jul. 2023 – Aug. 2024

- Originated the product concept from an idea developed through SOPT and helped build the founding team to launch **fitapat**, an AI-based pet customization service.
- Led product and app/web planning for an end-to-end service that used generative image AI to create personalized pet characters from user-provided photos and connect the generated content to customized merchandise production.
- Led marketing and brand operations through product launch; developed the product experience around an AI customization system capable of generating **80+ character concepts from**

eight pet photos, and the fitapat platform and its physical products were **showcased at CES 2024**.

LEADERSHIP & ACTIVITIES

SOPT (Shout Our Passion Together)
Nationwide University Student IT Venture & Startup Organization
Product Manager, 31st Cohort Jul. 2022 – Jan. 2023
Vice President, 32nd Cohort Jan. 2023 – Jul. 2023

- Served as Vice President of the 32nd Cohort and previously worked as a Product Manager in the 31st Cohort, where I planned **ZOOC**, a digital service for preserving memories between pets and their families.

Sookmyung Admissions Ambassador "Polaris," 16th Cohort
Sookmyung Women's University, Office of Admissions
Vice President Mar. 2021 – Feb. 2023

- Led university admissions and outreach activities, including prospective-student mentoring and admissions programs.

Statistics Education Talent Donation Activity
Statistics Korea (KOSTAT) — University Student Statistics Education Talent Donation Team
Volunteer Educator May 2021 – Dec. 2021

- Provided statistical education to elementary, middle, and high school students.

SOOK-TAT, Department of Statistics Data Analysis Society
Sookmyung Women's University, Department of Statistics
Member Sep. 2022 – Jul. 2023

- Participated in regular data analysis studies and team-based activities.

Department of Statistics Student Council
Sookmyung Women's University, Department of Statistics
Member Mar. 2020 – Dec. 2021

- Participated in regular meetings and organized departmental events and student activities.

KB Campus Star, 17th Cohort
KB Kookmin Bank
Student Ambassador May 2022 – Nov. 2022

- Created promotional content and participated in outreach activities for university students.

Amorepacific × Sookmyung Women's University A.I. Team Project
Amorepacific Foundation Oct. 28, 2024 – Dec. 6, 2024

- Completed a team project using generative AI to create **cosmetic concept boards and advertising video content**.

TECHNICAL SKILLS

Programming Languages: Python, R, SQL
ML & Frameworks: PyTorch, CUDA, LangChain, LangGraph, pandas, NumPy
Generative AI & LLM Systems: Diffusion Models, Transformer-based Models, LLM Tool Calling, MCP
Databases & Vector Search: Qdrant, PostgreSQL/pgvector, Neo4j
Simulation: MuJoCo
Developer Tools: Git, GitHub, Visual Studio Code, Overleaf
Certifications: ADsP (Advanced Data Analytics Semi-Professional)